

BACKTESTING FRAMEWORK

Backtest Report — 4-EMA Breakout Strategy

Quantitative Strategy Performance Verification & Trade Ledger

RUN IDENTIFICATION

test_run_large_3000_trades

GENERATION DATE

2026-09-24 16:33:38 UTC

BACKTEST PERIOD

2021-01-01 – 2021-01-10

UNIVERSE TRADED

1 symbols (NSE 101–750)

Executive Summary

Core Performance Metrics & Strategy Synthesis

Over the 0.1-year period, the strategy took 2500 trades, achieved a win rate of 100.0%, and grew capital from ₹500,000.00 to ₹500,050.00, a CAGR of +0.41%.

Final Capital ₹500,050.00	Total Return % +0.01%	Number of Trades 2500
Average Profit ₹50.00	Average Loss ₹0.00	Win % 100.0%

Advanced Ratio / Metric	Value	Standard Definition & Benchmark Context
CAGR (Compound Annual Growth Rate)	+0.41%	Annualized geometric return rate across simulation calendar window
Sharpe Ratio	11.22	Annualized excess return per unit of volatility (risk-free rate = 0%)
Sortino Ratio	0.00	Annualized return per unit of downside deviation (minimum acceptable return = 0%)
Max Drawdown	0.00% (₹0.00)	Maximum peak-to-trough equity reduction experienced during backtest
Calmar Ratio	0.00	Ratio of CAGR to Maximum Drawdown (absolute percentage)
Profit Factor	inf	Gross absolute profits divided by gross absolute losses
Expectancy	₹50.00	Average monetary gain/loss expected per executed round-trip trade
Avg Days Held	9.0 days	Average holding duration from session T+1 execution to trade exit

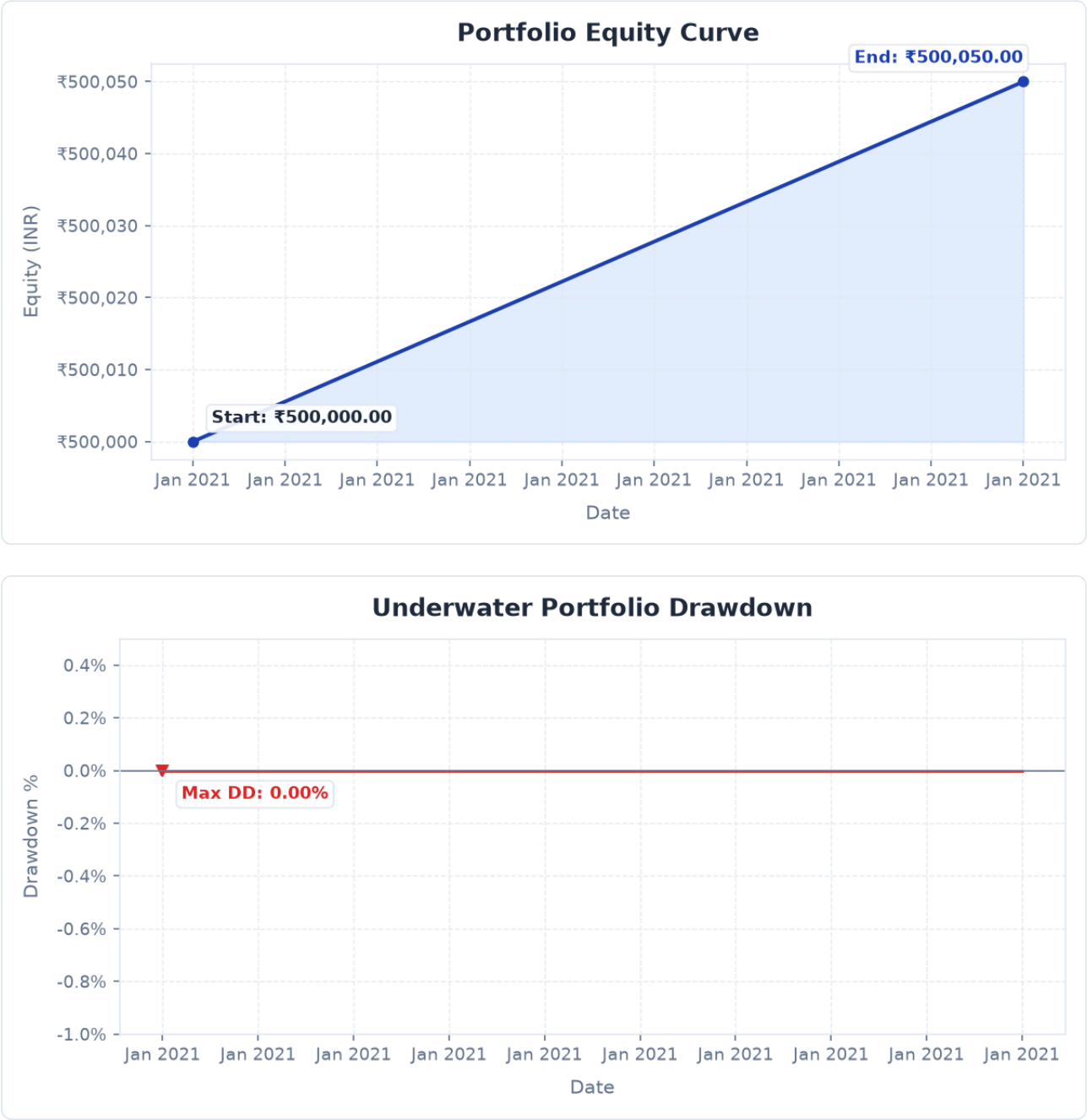
Strategy Configuration

Execution Parameters & Reproducibility Fingerprint

Configuration Parameter	Serialized Value
Data Snapshot Hash	7d6080be581234de2274c1a16897a74d4b6b539fd861ebbd8b87be6804fea76e
Git Commit SHA	a88c52954164c669ce70734cbe5820fcbd273e52

Performance Analytics & Drawdown

Mark-to-Market Equity Progression & Underwater Curves



Monthly Compounded Returns Matrix

Month x Year Compounded Return Distribution

Monthly Compounded Returns (%)												
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
2021	0.0%	—	—	—	—	—	—	—	—	—	—	—

Trade Log

Closed Round-Trip Transactions Ledger (2500 trades)

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Symbol	Entry Date	Entry Price	Qty	Exit Date	Exit Price	P&L (INR)	P&L (%)	Days	Exit Reason
ALPHA	2021-01-01	₹100.00	10	2021-01-10	₹105.00	+50.00	+5.00%	9	SIGNAL
ALPHA	2021-01-01	₹100.00	10	2021-01-10	₹105.00	+50.00	+5.00%	9	SIGNAL
ALPHA	2021-01-01	₹100.00	10	2021-01-10	₹105.00	+50.00	+5.00%	9	SIGNAL
ALPHA	2021-01-01	₹100.00	10	2021-01-10	₹105.00	+50.00	+5.00%	9	SIGNAL
ALPHA	2021-01-01	₹100.00	10	2021-01-10	₹105.00	+50.00	+5.00%	9	SIGNAL
ALPHA	2021-01-01	₹100.00	10	2021-01-10	₹105.00	+50.00	+5.00%	9	SIGNAL
ALPHA	2021-01-01	₹100.00	10	2021-01-10	₹105.00	+50.00	+5.00%	9	SIGNAL
ALPHA	2021-01-01	₹100.00	10	2021-01-10	₹105.00	+50.00	+5.00%	9	SIGNAL
ALPHA	2021-01-01	₹100.00	10	2021-01-10	₹105.00	+50.00	+5.00%	9	SIGNAL
ALPHA	2021-01-01	₹100.00	10	2021-01-10	₹105.00	+50.00	+5.00%	9	SIGNAL
ALPHA	2021-01-01	₹100.00	10	2021-01-10	₹105.00	+50.00	+5.00%	9	SIGNAL
ALPHA	2021-01-01	₹100.00	10	2021-01-10	₹105.00	+50.00	+5.00%	9	SIGNAL

Appendix: Methodology Assumptions & Audit Trail

Traceability Specifications & Environment Governance

Provenance & Reproducibility Audit Record

GIT COMMIT SHA

a88c52954164c669ce70734cbe5820fcbd273e52

STRATEGY CONFIG HASH (SHA-256)

44136fa355b3678a1146ad16f7e8649e94fb4fc21fe77e8310c060f61caaff8a

DATA SNAPSHOT HASH (SHA-256)

7d6080be581234de2274c1a16897a74d4b6b539fd861ebbd8b87be6804fea76e

RUNTIME ENVIRONMENT

Python 3.14.7, FastAPI 0.141.1, Pandas 3.0.6

Project Assumptions & Methodology Rules

This document records the foundational assumptions, methodology decisions, and edge-case behaviors locked for the EquiTest NSE quantitative backtesting platform.

0. PRD §4 Open Questions & Design Decisions (Consolidated)

The PRD defines several open questions regarding strategy execution, edge cases, and portfolio constraints. Below are the locked implementation choices and defaults:

Open Question #1: Universe Selection & Large-Cap Exclusion

- **PRD Query:** Should the strategy include large-cap stocks (Nifty 50, Nifty Next 50) or focus strictly on mid/small-caps?
- **Locked Choice:** The strategy strictly trades equities ranked **101 to 750 by market capitalization** (Nifty Midcap 150, Nifty Smallcap 250, Microcap 250). Large-caps (ranks 1 to 100) are explicitly excluded from the equity trading universe.
- **Rationale:** The 4-EMA trend following and 52-week breakout momentum strategy targets high-beta growth stocks rather than mature large-caps.
- **TODO:** PO confirm whether parameter sweeps across custom rank bands (e.g. 101–250 for Midcap only) should become strategy profiles in the UI.

Open Question #2: Stop Loss Execution on Overnight Gap-Downs

- **PRD Query:** If a stock gaps down overnight below the 7% stop loss price ($\text{Open}_T \leq \text{SL Price}$), at what price does the exit order fill?
- **Locked Choice:** The stop exit fills at the **Market Open price** ($\text{Exit Price} = \text{Open}_T \times 0.9990$), NOT at the stop loss price.
- **Rationale:** Modeling fills at the stop-loss price would represent unrealizable price improvement during market gap events. Realized losses exceed 7% ($\text{Loss} > 7\%$) and portfolio drawdown exceeds 2%, accurately capturing market gap risk.

Open Question #3: Candidate Ranking Rule Under Capital Constraints

- **PRD Query:** When multiple universe constituents trigger entry signals simultaneously and free portfolio capital cannot fund all candidates, what is the prioritization rule?
- **Locked Choice:** Candidates are ranked by **Momentum Breakout Score** descending: $\text{Score} = \frac{\text{Close}_T - \text{EMA20}_T}{\text{EMA20}_T} \times \text{descending}$
- **Rationale:** Prioritizes stocks whose closing prices have broken furthest above their 20-day exponential moving average, maximizing initial momentum. Ties are broken alphabetically by ticker symbol. Pluggable rankers (`alphabetical` , `52w_proximity`) are supported via `ranking_method` .
- **TODO:** PO confirm whether secondary tie-breaking should incorporate relative volume breakout metrics.

Open Question #4: Cash Equities Lot Sizing

- **PRD Query:** Should the engine enforce discrete share quantities or fractional shares, and should lot sizes default to 1?
- **Locked Choice:** `lot_size` defaults to **1 share** for NSE cash equities. Fractional shares are prohibited; quantities are floored to integer shares via `floor(Risk Amount / (Entry * 0.07))` .

- **TODO:** `PO confirm` whether lot sizing configuration should support derivatives basket multiples if F&O contracts are traded in future phases.

Open Question #5: Corporate Actions & Rolling 52-Week Ceiling

- **PRD Query:** How should splits, bonus issues, and dividends affect rolling 52-week highs and moving averages?
- **Locked Choice:** All calculations operate on **corporate action adjusted prices** (`adj_close`). The rolling 52W high is normalized by the historical corporate action adjustment ratio (`high * (adj_close / close)`), ensuring splits do not distort historical price ceilings.

Open Question #6: Friction & Slippage Cost Modeling

- **PRD Query:** What transaction costs and slippages should be assumed for mid/small-cap NSE equities?
- **Locked Choice:** A fixed **10 basis points (0.10%) per side** ($\$0.20\%$ round trip) is applied to all market orders ($\text{Buy} = \text{Price} \times 1.0010\%$, $\text{Sell} = \text{Price} \times 0.9990\%$).
- **TODO:** `PO confirm` whether exchange turnover charges, STT (Securities Transaction Tax), GST, and stamp duty should be itemized separately from execution slippage.

1. Trading Universe & Bias Prevention

1. **Trading Universe Constituents (NSE 101–750):**
2. **Exclusion of Top 100:** The strategy strictly trades stocks ranked 101 to 750 by market capitalization (Nifty Midcap 150, Nifty Smallcap 250, and Microcap 250). Large-cap stocks (ranks 1 to 100) are explicitly excluded from the equity trading universe.
3. **Benchmark Index:** NIFTY 50 (`^NSEI / NIFTY50`) benchmark is ingested and tracked separately solely for the market regime filter (`NIFTY > EMA 50` and `NIFTY > EMA 200`) and does not enter the stock trading universe.
4. **Survivorship Bias Handling:** Point-in-time constituent lists are resolved via `/data/fixtures/constituents.parquet` or `universe_membership` database records. If point-in-time data is missing for historical dates, the current constituent list is used as fallback, and `survivorship_bias: true` is explicitly flagged.

2. Technical Indicators & Mathematical Formulas

1. **Exponential Moving Averages (EMAs):**
2. **Formula:** Vectorized calculation using pandas `.ewm(span=span, adjust=False, min_periods=span).mean()` with smoothing factor $\alpha = \frac{2}{\text{span} + 1}$.
3. **Price Input:** All equity EMAs are computed on corporate-action adjusted close (`adj_close`).
4. **Warm-up Window:** Initial $(\text{span} - 1)$ sessions produce `NaN` (or `null` in JSON responses). The first valid non-`NaN` value begins exactly at row `span` (e.g., EMA-200 begins at row 200).
5. **Accuracy:** Verified to match standard technical analysis libraries (TA-Lib) within numerical tolerance of 10^{-6} .
6. **52-Week Rolling High:**
7. **Lookback Window:** 252 trading sessions (approx. 52 trading weeks in Indian equity markets).
8. **No Look-Ahead Guarantee:** Implemented as `high.shift(1).rolling(window=252, min_periods=252).max().Bar` `N`'s 52W high equals $\max(\text{high}[N-252:N])$ and strictly excludes bar `N`'s own high.
9. **Corporate Action Continuity:** If a stock underwent splits or bonus issues, the high series is normalized by the corporate adjustment factor (`high * (adj_close / close)`) to prevent distorted historical highs from invalidating breakout signals.

3. Order Execution & Look-Ahead Prevention

1. **Signal Generation & Order Execution:**
2. Indicator values and trading signals are evaluated on session `T`'s closing prices.
3. Any trade triggered on session `T` executes on session `T+1` at the **Market Open price**.
4. No intraday look-ahead is permitted in end-of-day backtesting.
5. **Price Filters:**
6. The 52W high filter requires `Price > 0.85 * 52W High` (i.e. within 15% of the 52-week high).

4. Signal Rules & Filter Invariants (Phase 3)

1. **Market Regime Filter (REQ-3.1):**
2. **Condition:** Evaluated on NIFTY 50 closing price: $\text{NIFTY_Close} > \text{EMA } 50 \text{ AND } \text{NIFTY_Close} > \text{EMA } 200$.
3. **Zero Trade Invariant:** If market regime evaluates to False on session T , all entry signals across the entire universe are rejected ($\text{entry_signal} = \text{False}$).
4. **Lookback Warm-up:** Before bar 200 of the benchmark series, regime evaluates to `False`.
5. **Stock Trend Filter (REQ-3.2):**
6. **Condition:** Evaluated on equity adjusted close: $\text{EMA } 20 > \text{EMA } 50 > \text{EMA } 150 > \text{EMA } 200$.
7. **Strict Hierarchy:** All four EMAs must be strictly stacked. If any adjacent EMAs are equal or inverted, trend evaluates to `False`.
8. **52-Week High Proximity Filter (REQ-3.3):**
9. **Condition:** $\text{Close} > 0.85 \times 52\text{W High}$ (strictly within 15% of the rolling 252-day peak).
10. **Scale Consistency:** Both the price and 52W high series use corporate-action adjusted pricing.
11. **Entry Trigger & Crossover Invariant (REQ-3.4):**
12. **Formula:** $(\text{Regime} \ \& \ \text{Trend} \ \& \ \text{Near}52\text{W}) \text{ on } T \text{ AND } (\text{Close}_T > \text{EMA}20_T) \text{ AND } (\text{Close}_{T-1} < \text{EMA}20_{T-1})$.
13. **Strict Crossover:** By default, bar $T-1$ requires $\text{Close} < \text{EMA}20$.
14. **TODO:** PO confirm whether exact equality at bar $T-1$ ($\text{Close}(T-1) == \text{EMA}20(T-1)$) should trigger an entry signal. Parameterized via `StrategyConfig.allow_crossover_equal = False`.
15. **Exit Trigger Invariant (REQ-3.5):**
16. **Formula:** $\text{Close}(T) < \text{EMA}20(T)$.
17. **Execution:** Signal generated on bar T close; execution occurs on session $T+1$ at Market Open.

5. Risk Management, Sizing & Stop Loss Invariants (Phase 4)

1. **Initial Stop Loss Calculation (REQ-4.1):**
2. **Formula:** $\text{SL Price} = \text{Entry Price} \times (1 - 0.07)$ (7% hard stop loss).
3. **Execution:** Evaluated on subsequent trading sessions. If $\text{Low}_T \leq \text{SL Price}$, a stop loss exit is triggered.
4. **Dynamic Account Risk (REQ-4.2):**
5. **Formula:** $\text{Risk Amount} = \text{Current Corpus} \times 0.02$ (2% risk per trade).
6. **Dynamic Adjustment:** Initial risk on ₹5,00,000 starting corpus is ₹10,000. Compounding increases dollar risk during equity peaks and reduces dollar risk during portfolio drawdowns.
7. **Position Sizing Derivation (REQ-4.3):**
8. **Formula:**
$$\begin{aligned} \text{Raw Qty} &= \frac{\text{Corpus} \times \text{Risk \%}}{\text{Entry Price} \times \text{SL \%}} \\ \text{Quantity} &= \left\lfloor \frac{\text{Raw Qty}}{\text{Lot Size}} \right\rfloor \times \text{Lot Size} \end{aligned}$$
9. **PRD Invariant & Arithmetic Relationship:**
 - At ₹5,00,000 corpus and ₹100 entry: Risk is ₹10,000, SL distance is ₹7.00 $\implies \text{Qty} = \lfloor 10000 / 7 \rfloor = 1,428$ shares, requiring ₹1,42,800 capital ($\approx 28.56\%$ allocation).
 - At ₹1,00,000 corpus and ₹100 entry: Risk is ₹2,000, SL distance is ₹7.00 $\implies \text{Qty} = \lfloor 2000 / 7 \rfloor = 285$ shares, requiring ₹28,500 capital (28.5% allocation).
 - **TODO:** PO confirm whether lot sizes for cash equities default strictly to 1 or should support index derivative/basket lot sizes. Currently defaulted to `lot_size = 1`.
10. **Overnight Gap-Down Semantics (Explicit Assumption):**
11. If a stock opens below the stop loss price ($\text{Open}_T \leq \text{SL Price}$), the stop exit executes at the **Open price** rather than the stop loss price.
12. **Impact:** Realized trade loss exceeds 7% ($\text{Loss} > 7\%$) and portfolio loss exceeds 2% ($\text{Drawdown} > 2\%$). The backtesting engine records realistic market gaps without artificial price improvement.
13. **Transaction Frictions & Slippage (REQ-0.3):**

14. Modeled at **10 basis points (0.10%) per side**:

- Buy execution: $\text{Price} \times 1.0010$
- Sell execution: $\text{Price} \times 0.9990$

15. **Portfolio Exposure & Capital Allocation**:

16. System enforces total portfolio exposure $\leq \text{Corpus}$.

17. Because each trade requires $\sim 28.5\%$ of capital, the portfolio naturally caps at a maximum of **3 to 4 concurrent positions**. New entry signals are rejected if $\text{Open Positions Value} + \text{Required Capital} > \text{Corpus}$.

6. Backtest Simulation Engine & Capital Constraints (Phase 5)

1. Signal Clashing & Ranking Rule (PRD Open Question #3 / REQ-5.3):

- When multiple universe stocks generate an entry signal on the same session $T-1$, candidates are ranked by momentum score:
$$\text{Score} = \frac{\text{Close}_{T-1} - \text{EMA20}_{T-1}}{\text{EMA20}_{T-1}} \propto \text{(descending)}$$
- `DefaultRanker` prioritizes stocks whose closing prices have broken furthest above their 20-day EMA. Pluggable `Ranker` protocol allows custom ranking functions.
- TODO: `PO confirm` whether tie-breakers should incorporate volume breakout ratio or liquidity filters beyond symbol alphabetical sorting.

5. Capital Constraint & Position Rejections (REQ-5.3):

- Total exposure constraint enforces `sum(open positions value) + required capital <= corpus`.
- On ₹5,00,000 starting corpus with $\sim 28.5\%$ position allocation, a maximum of 3 concurrent positions can be opened. The 4th concurrent signal on the same session is rejected with `reason: "insufficient_capital"` and logged in `BacktestResult.rejections`.

8. Execution Semantics & Ordering:

- On each session T :
 - Exits are evaluated and executed first at Market Open (SL hit, gap-down, or $T-1$ exit signal).
 - Free capital is updated.
 - Entry candidates from $T-1$ signals are ranked and executed at Market Open with buy slippage until free capital is exhausted.
 - Portfolio equity is marked-to-market using session T closing prices.

7. Parameter Sweeps, Scenario Testing & Versioning (Phase 6)

1. Config Schema & Backward Compatibility (REQ-6.1):

- `StrategyConfig` is implemented as a Pydantic v2 `BaseModel` supporting runtime validation (`gt=0` , `lt=1` , etc.), dictionary serialization (`to_dict()`), and legacy alias compatibility (`corpus` $\rightarrow$ `capital` , `stop_loss_pct` $\rightarrow$ `sl_pct` , `ema_trend_spans` $\rightarrow$ `ema_spans` , `regime_spans` $\rightarrow$ `regime_ema_spans`).
- Percentage parameters (`sl_pct` , `risk_pct`) automatically normalize whole numbers ≥ 1.0 (e.g. $\$5 \rightarrow 0.05$, $\$2 \rightarrow 0.02$) to fractional decimals for seamless user input.

4. Pluggable Ranking Protocols:

5. Supported candidate prioritization rules:

- `"momentum"` : Sort by breakout percentage above EMA-20: $(\text{Close}_{T-1} - \text{EMA20}) / \text{EMA20}$ descending.
- `"alphabetical"` : Deterministic lexical sort by ticker symbol ascending.
- `"52w_proximity"` : Proximity to 52-week rolling peak: $\text{Close}_{T-1} / \text{High52W}_{T-1}$ descending.

6. Universe Rank Parameterization:

- `universe_start_rank` (default 101) and `universe_end_rank` (default 750) allow parameter sweeps across market-cap segments (e.g., Midcap 150: 101–250, Smallcap 250: 251–500, Microcap 250: 501–750).

8. Cartesian Sweep Limits & Execution Semantics (REQ-6.2):

- Parameter sweeps generate the full Cartesian product across candidate parameter lists.
- Grid size is capped at 50 permutations per request to prevent worker saturation.

11. Sweep runs execute asynchronously in background tasks, reporting live status (`pending` , `running` , `partial` , `completed` , `failed`) and per-child progress metrics.
12. **Multi-Run Metric Alignment & Overlaid Curves (REQ-6.3):**
13. Comparison endpoint (`GET /api/v1/backtest/compare?run_ids=...`) returns aligned performance metrics and synchronized mark-to-market equity curves across multiple runs for side-by-side evaluation.
14. **Annualized Returns & Config Versioning:**
15. All completed runs record `config_version="1.0"` , optional parent `sweep_id` , `cagr` computed as $\left(\frac{\text{Final Capital}}{\text{Initial Capital}}\right)^{\frac{1}{252N}} - 1$ \$, and `max_drawdown_pct` .

8. Reporting, Performance Metrics & Exports (Phase 7)

1. **Performance Metric Mathematical Specifications (REQ-7.1):**
2. **Annualization Factor:** 252 trading sessions per calendar year for Indian markets.
3. **CAGR:** $\left(\frac{\text{Final Capital}}{\text{Initial Capital}}\right)^{\frac{1}{252N}} - 1$ \$, where N is total trading sessions in the equity curve.
4. **Max Drawdown:** Peak-to-trough decline on daily marked-to-market equity. Both percentage (`max_drawdown_pct` , 4 decimal places) and absolute rupee loss (`max_drawdown_amount` , 2 decimal places) are calculated.
5. **Sharpe Ratio:** Risk-free rate $r_f = 0.0$ \$. Annualized as $\frac{\mu_r}{\sigma_r} \times \sqrt{252}$ \$, matching `empyrical.sharpe_ratio(..., risk_free=0, period="daily")` to within 10^{-6} \$.
6. **Sortino Ratio:** Minimum Acceptable Return $\text{MAR} = 0.0$ \$, annualization 252. Downside deviation is computed over all return observations with positive returns clipped to 0: $\sigma_d = \sqrt{\frac{1}{N} \sum \min(0, r_i)^2}$ \$, matching `empyrical.sortino_ratio(..., required_return=0, period="daily")` to within 10^{-6} \$.
7. **Calmar Ratio:** $\frac{\text{CAGR}}{|\text{Max Drawdown \%}|}$ \$. If drawdown is zero or CAGR is negative, Calmar is set to 0.0.
8. **Profit Factor:** $\frac{\sum \text{Gross Profits}}{\sum |\text{Gross Losses}|}$ \$. If gross losses are zero, returns ∞ \$ (`float('inf')`), formatted as " ∞ " in UI and "Infinity" in JSON.
9. **Expectancy:** Monetary average return per closed trade: $(\text{Win Rate} \times \text{Avg Profit}) - ((1 - \text{Win Rate}) \times |\text{Avg Loss}|)$ \$ in INR.
10. **Avg Days Held:** Arithmetic mean of duration in calendar days from entry to exit across all closed trades (1 decimal place).
11. **Decimal Serialization:** All float metrics support conversion to Python `Decimal` via `to_decimal_dict()` with precision matching documented requirements (2 decimal places for INR, 4 for rates/ratios, 1 for days).
12. **Monthly Returns Aggregation Matrix (REQ-7.2):**
13. Aggregated directly from daily marked-to-market portfolio equity curves.
14. Monthly return is compounded as $\frac{\text{Equity}_{\text{month_end}}}{\text{Equity}_{\text{month_start}}} - 1.0$ \$.
15. Missing months prior to the backtest start or after the backtest end are represented as `null` , avoiding artificial zero-return distortion.
16. Annual return is compounded across available trading months: $\prod (1 + r_m) - 1.0$ \$.
17. **Multi-Format Export Engine (REQ-7.3):**
18. **CSV Export:** Trade ledger table (`{run_id}_trades.csv`).
19. **XLSX Export:** Multi-worksheet OpenXML workbook (`Executive Summary` , `Performance Metrics` , `Trade Ledger` , `Monthly Returns`). Implemented using standard library `zipfile` and `SpreadsheetML` XML, requiring zero third-party C-extensions or external dependencies.
20. **ZIP Export:** Complete archive containing `metrics.csv` , `trades.csv` , `monthly_returns.csv` , and `equity_curve.csv` .